

An Investigation into QFT Digitisation Methods for Quantum Computers

Why?

- Noisy Intermediate-Scale Quantum era (NISQ Era)
- QFT
- Other Errors (Digitisation)

What's already been done?

[1] Digitization of Scalar Fields for Quantum Computing by Klco and Savage

More Foundational Works:

[2] Quantum Algorithms for Quantum Field Theories by Jordan, Lee and Preskill

[3] Simulating collider physics on quantum computers using effective field theories by Bauer and Nachman

[4] Quantum simulations of one dimensional quantum systems by Somma

Errors

Isolate digitisation errors

- Use known ground state energies

$$\bar{H} = \frac{1}{2}\bar{\Pi}^2 + \frac{1}{2}\bar{\phi}^2$$

$$\bar{H} = \frac{1}{2}\bar{\Pi}^2 + \frac{1}{2}\bar{\phi}^2 + \frac{\bar{\lambda}_0}{4!}\bar{\phi}^4$$

Qubits

For a given Quantum Computer:

$$n_{\phi} = 2^{n_Q}$$

where n_{ϕ} is the number of possible states able to be represented by a number of qubits n_Q

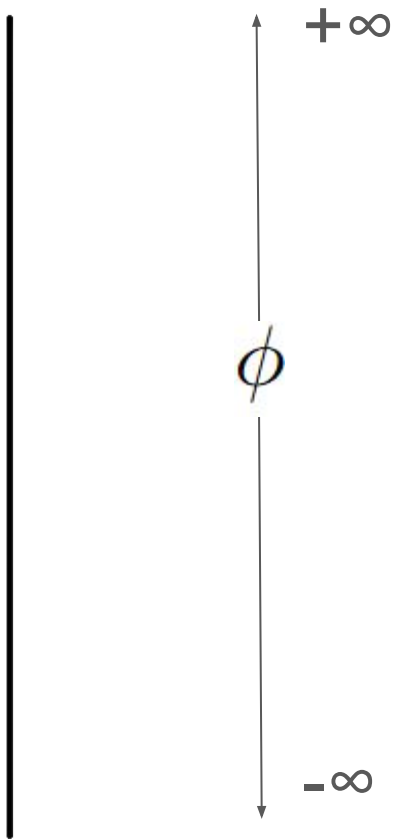
3 Methods of Digitisation

Jordan-Lee-Preskill

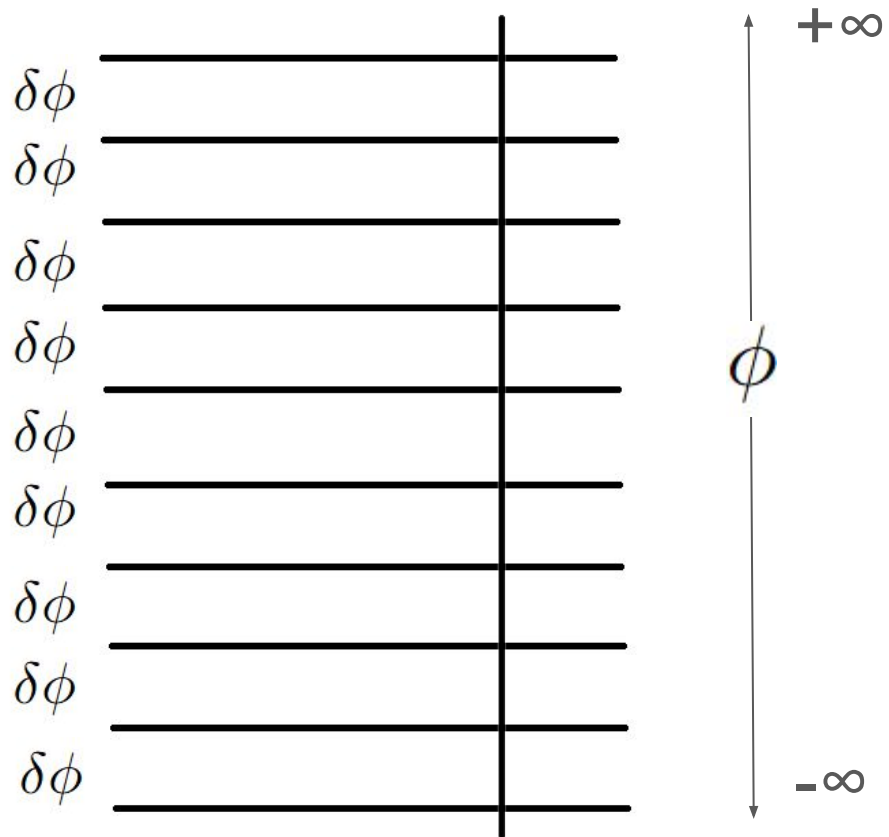
Finite Difference

Harmonic Oscillator Basis

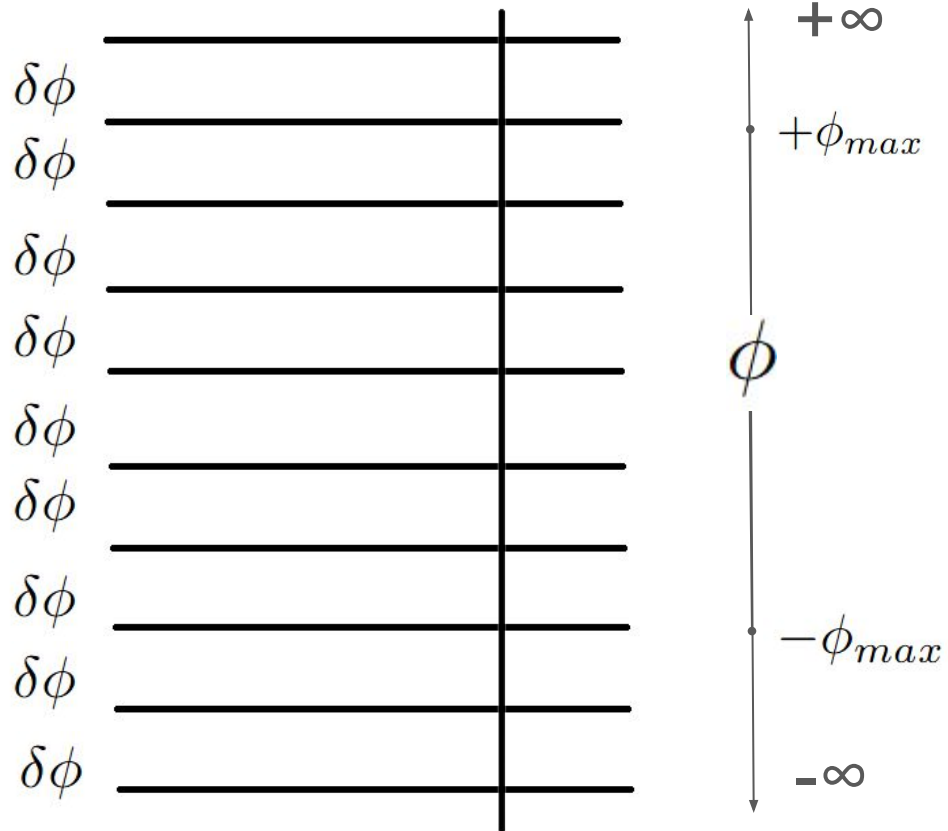
Jordan-Lee-Preskill



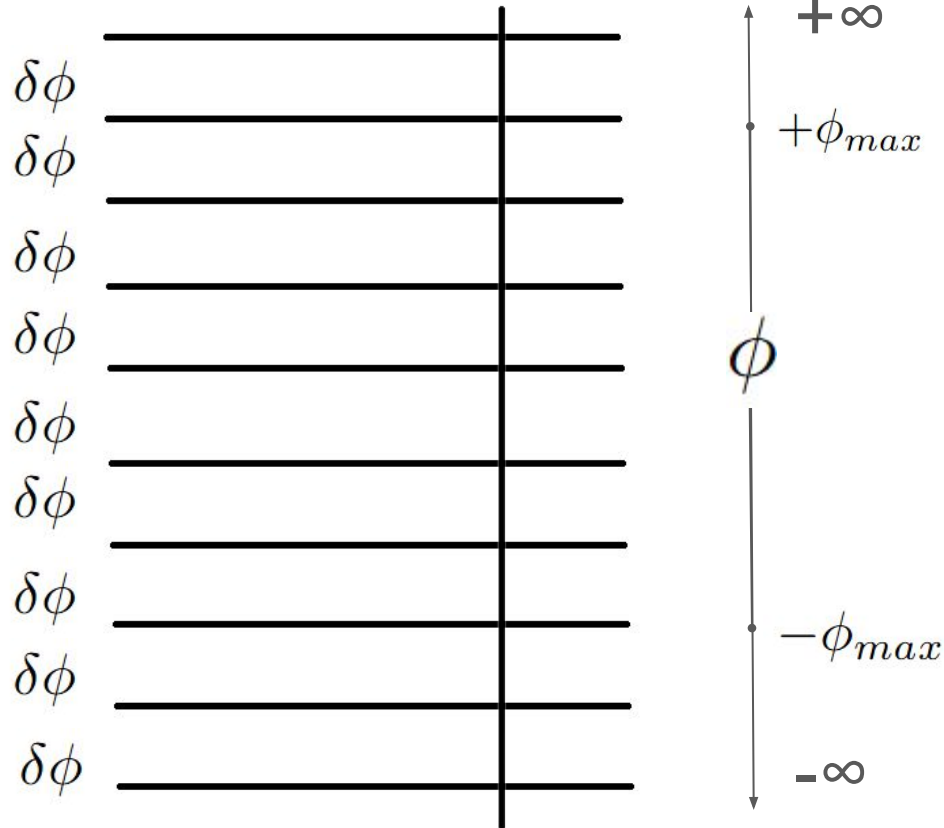
Jordan-Lee-Preskill



Jordan-Lee-Preskill

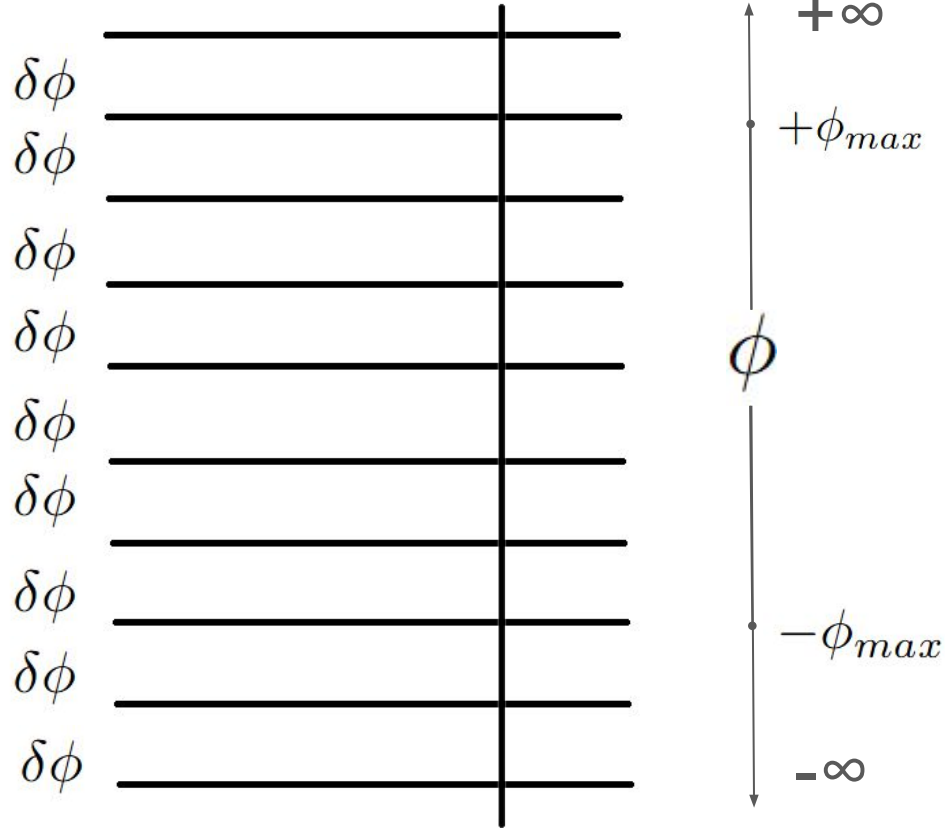


Jordan-Lee-Preskill



$$\delta\phi = \frac{2\phi_{max}}{n_\phi - 1}$$

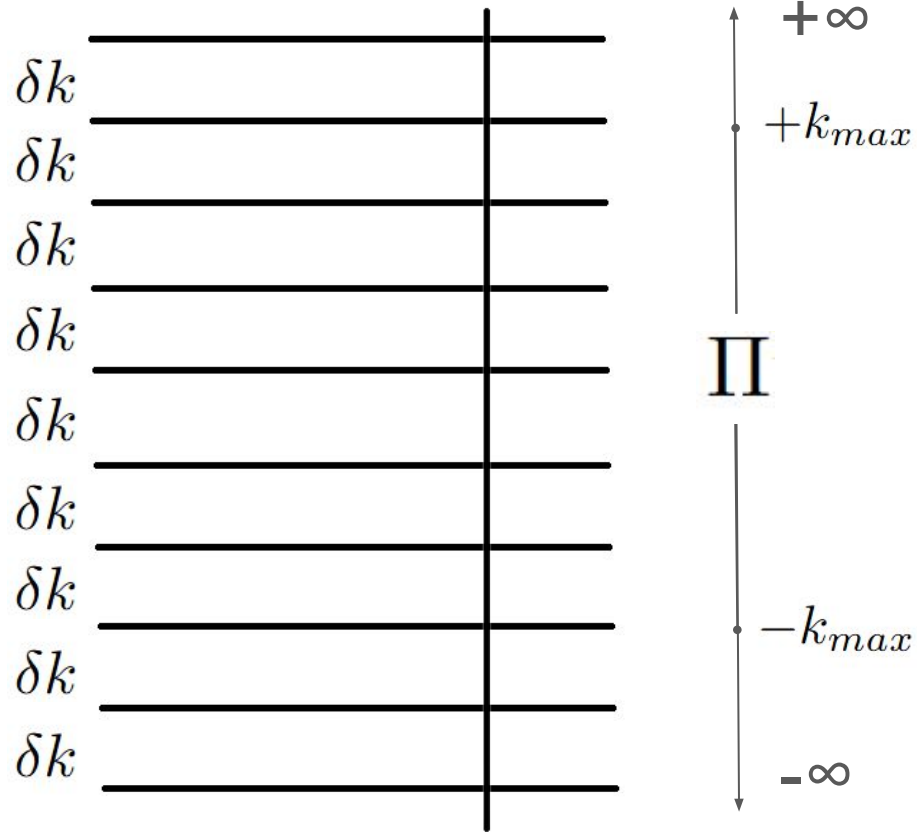
Jordan-Lee-Preskill



$$\bar{H} = \frac{1}{2} \bar{\Pi}^2 + \frac{1}{2} \bar{\phi}^2$$

(constructed as diagonal matrix of values of ϕ)

Jordan-Lee-Preskill



$$k_{max} = \frac{\pi}{\delta\phi}$$

$$\delta k = \frac{2k_{max}}{n_\phi}$$

Jordan-Lee-Preskill

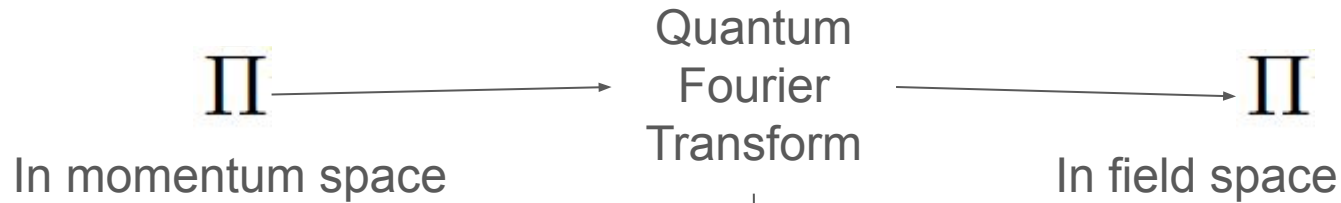


$$k_{max} = \frac{\pi}{\delta\phi}$$

$$\delta k = \frac{2k_{max}}{n_\phi}$$

Still in Momentum Space

Jordan-Lee-Preskill



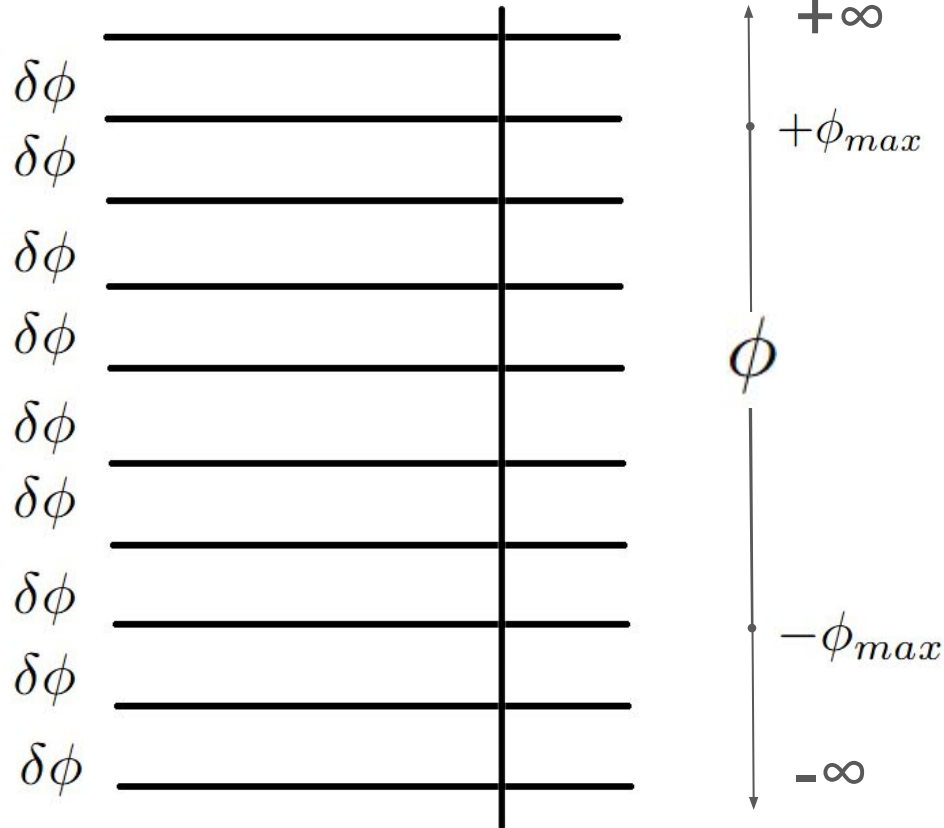
(make a unitary fourier matrix using our ϕ matrix which transforms between field and momentum space) [1]

Jordan-Lee-Preskill

$$\bar{H} = \frac{1}{2}\bar{\Pi}^2 + \frac{1}{2}\bar{\phi}^2$$
Two large green checkmarks are positioned below the terms $\bar{\Pi}^2$ and $\bar{\phi}^2$ in the equation, indicating that the expression is correct or verified.

Finite Difference (FD)

Finite Difference (FD)



$$\delta\phi = \frac{2\phi_{max}}{n_{\phi} - 1}$$

Finite Difference (FD)

$$\bar{\pi}^2 = -\frac{\partial^2}{\partial\phi^2}$$

Operator

$$\psi''(\phi) = \frac{\psi(\phi - \delta\phi) + \psi(\phi + \delta\phi) - 2\psi(\phi)}{\delta\phi^2}$$

$$= \frac{\psi(\phi_{n-1}) + \psi(\phi_{n+1}) - 2\psi(\phi_n)}{\delta\phi^2}$$

Finite Difference
Approximation

$$\bar{\Pi}^2 = \langle\psi_n|\bar{\pi}^2|\psi_m\rangle = \langle\psi_n|-\psi_{m-1}-\psi_{m+1}+2\psi_m\rangle$$

Combination

Finite Difference (FD)

$$\bar{H} = \frac{1}{2}\bar{\Pi}^2 + \frac{1}{2}\bar{\phi}^2$$
Two large green checkmarks are positioned below the terms $\bar{\Pi}^2$ and $\bar{\phi}^2$ in the equation, indicating that the finite difference approximation is correct for these terms.

Finite Difference Corrected (FDC)

$$\bar{H} = \frac{1}{2}\bar{\Pi}^2 + \frac{1}{2}\bar{\phi}^2 + \frac{1}{24}\delta\phi^2\bar{\Pi}^4$$

Harmonic Oscillator Basis

We choose to work in a truncated set of harmonic oscillator eigenstates:

$$\{|0\rangle, |1\rangle, \dots, |n_\phi - 1\rangle\}$$

Assume: true low energy states of our hamiltonian can be approximated by a linear combination of these states

Harmonic Oscillator Basis

How do we express $\bar{\Pi}$ and $\bar{\phi}$ in terms of these states?

Harmonic Oscillator Basis

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$$H_{HO} = \frac{1}{2}\Pi^2 + \frac{1}{2}\omega^2\phi^2$$

Harmonic Oscillator Basis

How do we express $\bar{\Pi}$ and $\bar{\phi}$ in terms of these states?

$$H_{HO} = \frac{1}{2}\Pi^2 + \frac{1}{2}\omega^2\phi^2$$

$$H_{HO} = \omega(a^\dagger a + \frac{1}{2})$$

Harmonic Oscillator Basis

$$a |n\rangle = \sqrt{n} |n-1\rangle$$

$$a^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle$$

Harmonic Oscillator Basis

$$a |n\rangle = \sqrt{n} |n-1\rangle$$

$$a^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle$$

$$\bar{\phi} = \frac{a + a^\dagger}{\sqrt{2\omega}}$$

$$\bar{\Pi} = i\sqrt{\frac{\omega}{2}}(a^\dagger - a)$$

Harmonic Oscillator Basis

$$\bar{H} = \frac{1}{2}\bar{\Pi}^2 + \frac{1}{2}\bar{\phi}^2$$
Two large green checkmarks are positioned below the terms $\bar{\Pi}^2$ and $\bar{\phi}^2$ in the equation, indicating that the variables are correctly identified or the equation is correct.

Nyquist-Shannon Sampling Theorem

- Variables are related to each other
- End up with a balancing issue

JLP (/FDC):

$$k_{max} = \frac{\pi}{\delta\phi}$$

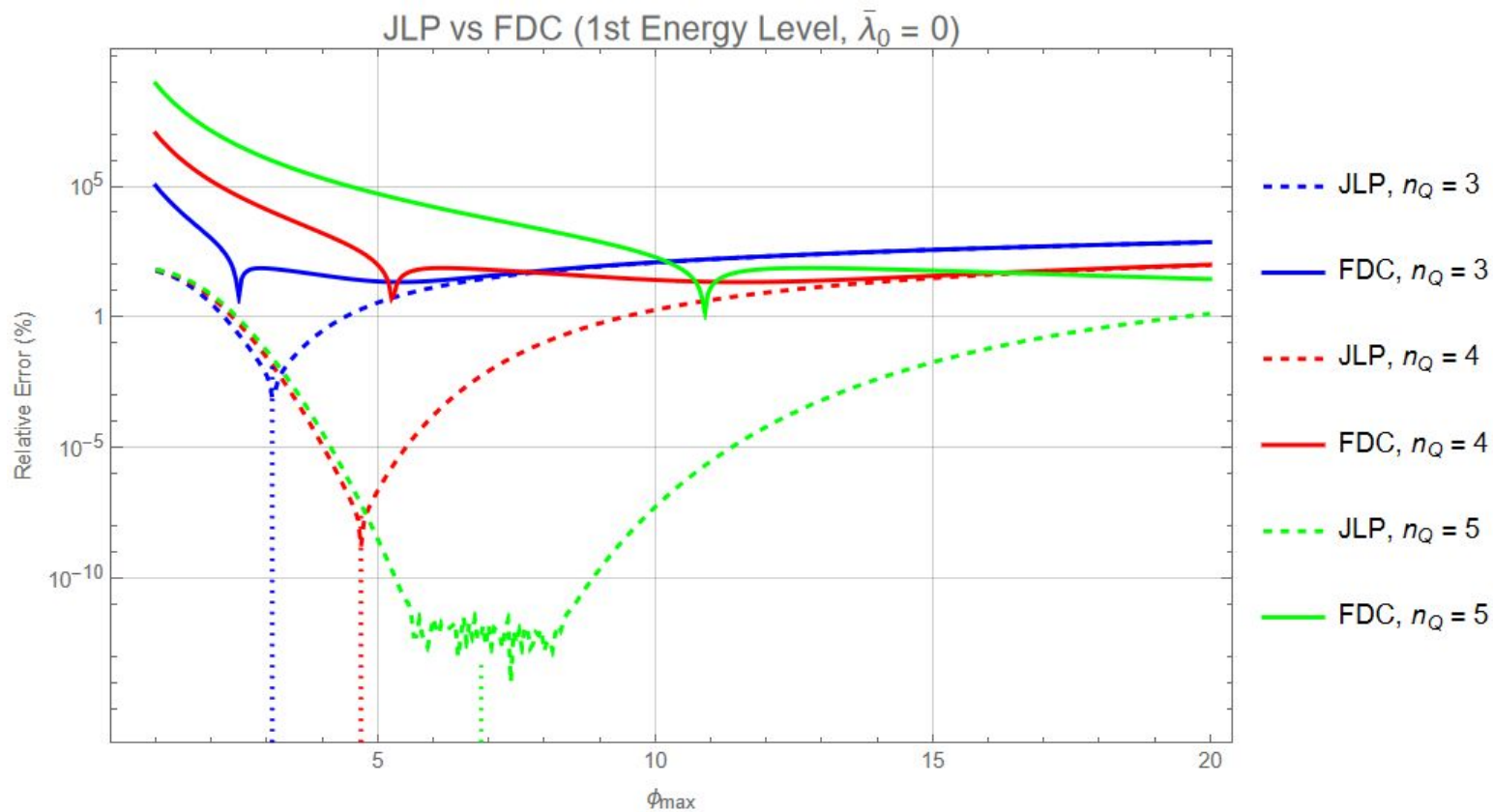
$$\phi_{max} = \frac{1}{\sqrt{\omega}} \sqrt{\frac{\pi(n_{\phi} - 1)^2}{2n_{\phi}}}$$

HO Basis:

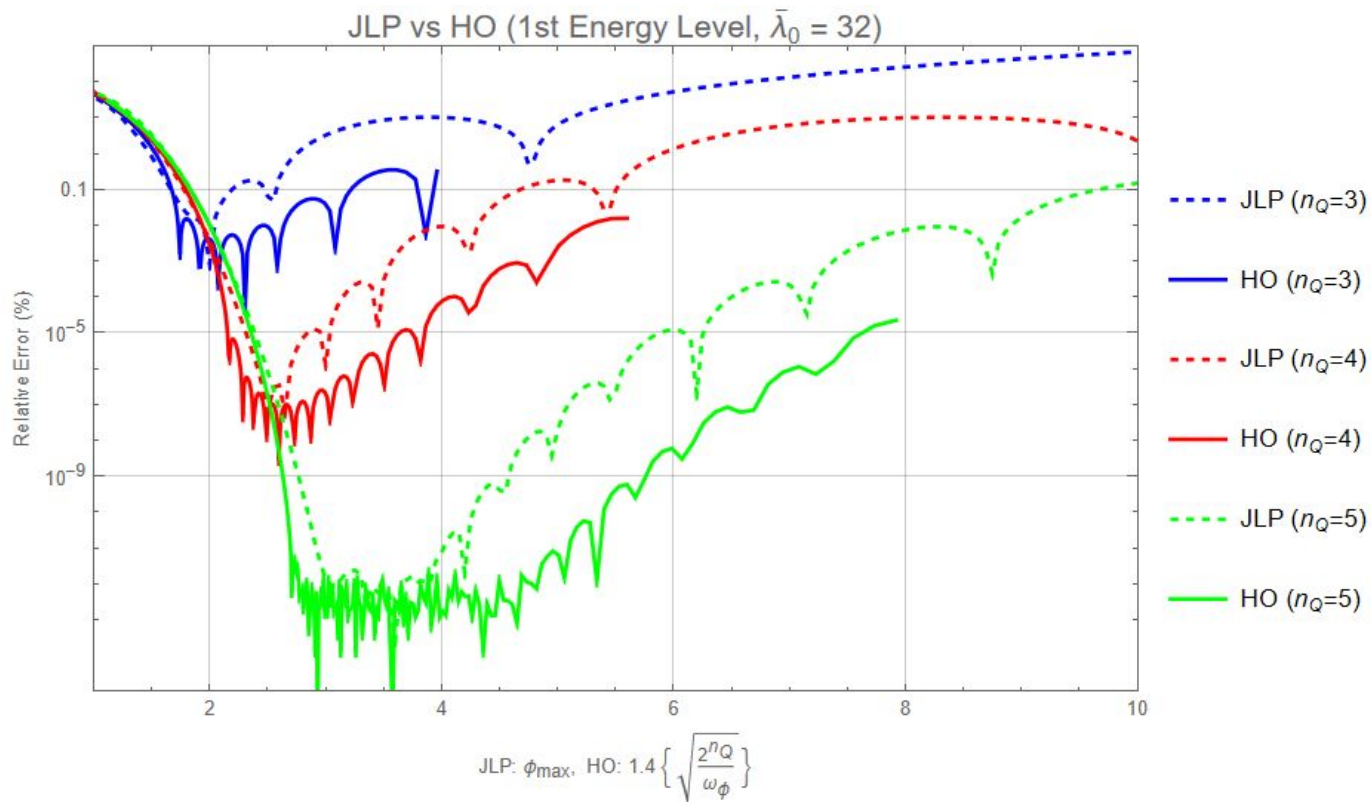
$$\bar{\phi} = \frac{a + a^{\dagger}}{\sqrt{2\omega}}$$

$$\bar{\Pi} = i\sqrt{\frac{\omega}{2}}(a^{\dagger} - a)$$

Results



Results



Looking forward, HTET what is it?

[5] - Hamiltonian Truncation Effective Theory by Cohen, Farnsworth, Houtz, Luty